Design session:

Simulation of hydrogen combustion with Cantera

Design Project Report

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Nomenclature

Latin Letters

P	Pressure	atm
P_0	Initial pressure	atm
T	Temperature	K
T_0	Initial temperature	K
t	time	s
t_{end}	End time	s
P	Pressure	atm
P_0	Initial pressure	atm
S_L	Laminar flame speed	m/s
$X_{[.]}$	Molar fraction	-

Acronyms / Abbreviations

0D, 1D, 2D, 3D	Zero-, One-, two-, three-dimensional
CH_4	Methane
H_2	Hydrogen
N_2	Nitrogen
O_2	Oxygen

Greek Symbols

τ_{ign}	Ignition delay	s
τ_r	Resident time	s
ϕ	Equivalent ratio	-

Introduction

Hydrogen is widely regarded as a promising fuel for future propulsion systems, thanks to its high calorific value and the absence of direct CO_2 emissions during combustion. However, hydrogen combustion exhibits specific kinetic and transport features that strongly affect auto-ignition behavior, laminar flame speeds, and the formation of pollutants such as nitrogen oxides (NOx). Understanding and modeling these fundamental phenomena is therefore essential for designing combustion systems that are both efficient and compliant with increasingly stringent environmental constraints.

In this design project, the combustion properties of hydrogen and hydrogen–methane mixtures are investigated using Cantera, a numerical library that solves detailed chemical kinetics coupled with thermodynamics and transport models. A first part focuses on the analysis of ignition delay as a function of initial temperature and pressure, in order to obtain the so-called "Z-curve" for hydrogen–oxygen mixtures and to explain its physico-chemical origin. The second part aims at computing the laminar flame speed of a stoichiometric H_2–O_2 flame, and then of an H_2/CH_4 blend as a function of the hydrogen fraction, which allows quantifying the influence of hydrogen addition on premixed flame reactivity.

Finally, a third part of the study deals with the modeling of a Perfectly Stirred Reactor (PSR) fed with a hydrogen–oxygen mixture, to predict NOx emissions as functions of residence time and equivalence ratio. All the questions in the assignment are addressed using Python–Cantera simulations, and the results are presented in a concise report limited to ten pages with figures, accompanied by the commented Python source files that constitute the second deliverable of the project.

1 Ignition delay

In our report, we used two methods to calculate the ignition delay for the combustion of hydrogen with air.

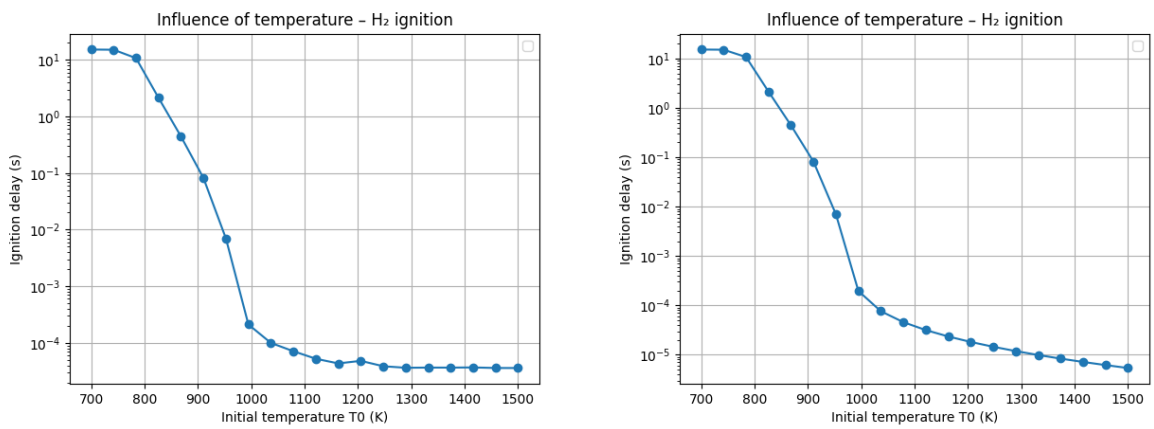
Method 1: OH-based method This method is often used in Cantera examples in shock tubes and in experimental work, where the goal is to deduce the ignition time marked by the maximum OH signal. To do this, simply define the ignition time as the moment when the mass fraction of the OH radical reaches its maximum value.

$$\tau_{ign,OH} = arg\max Y_{OH}(t) \tag{1}$$

Method 2: T Gradient based In this method, we will use a gradient-based criterion based on temperature (sometimes referred to as heat flux, but in practice, this is closely related to temperature, which amounts to the same thing). It could also be pressure or the mass fraction of OH. Here, the ignition delay is therefore defined as the moment when $\frac{dT}{dt}$ is at its maximum.

$$\tau_{ign,T} = arg\max \frac{dT}{dt}(t) \tag{2}$$

1.1 Ignition delay VS Temperature



(a) OH based method

(b) Gradient based method

Figure 1: Ignition delay as a function of initial temperature for a fixed pressure and equivalence ratio.

The ignition-delay curve obtained for the H_2/O_2 mixture shows three temperature regimes that explain why it is called a Z-curve. At low initial temperatures (around 700–800 K), chain-branching reactions are weak and radical-termination reactions dominate, so radicals accumulate slowly, and the ignition delay is long and only weakly dependent on temperature, forming an upper, almost horizontal branch. As the temperature increases to an intermediate range (about 800–1000 K), chain-branching chemistry becomes much more effective, which

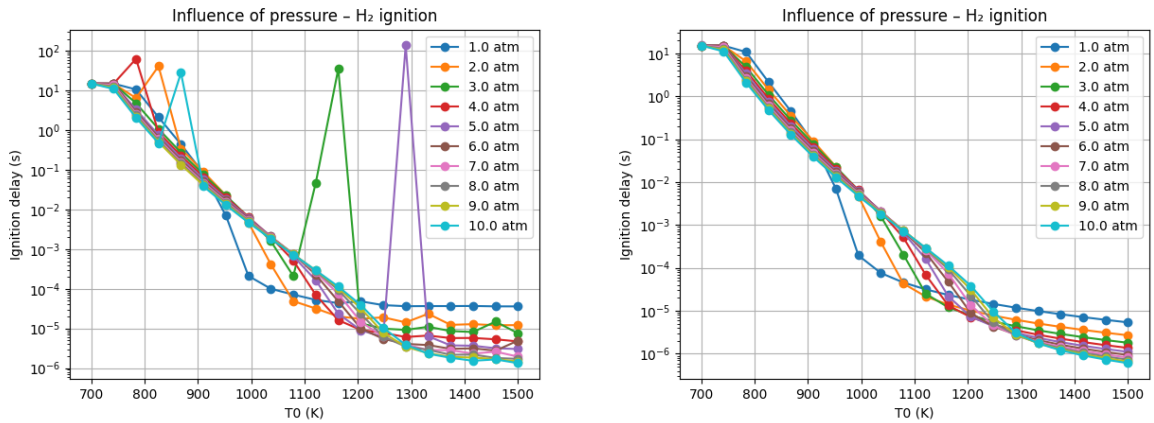
strongly accelerates radical production and causes the ignition delay to drop by several orders of magnitude, giving the steep central part of the Z.

At higher initial temperatures (typically above 1000–1100 K), most key reactions are already very fast, so further temperature increases have a more limited effect on the overall kinetics: the ignition delay still decreases, but much more moderately, producing a lower, nearly horizontal branch at short delay times. Plotted on a semi-logarithmic scale, the succession of these three regimes—long delays at low temperature, a sharp transition at intermediate temperature, and short delays at high temperature naturally forms a curve that resembles the letter "Z", hence the name Z-curve.

However, we observe a difference between the values despite the same curve shape between the two methods, which can be explained by the fact that the gradient-based method detects the moment of maximum temperature rise, which is slightly earlier than the OH maximum, leading to shorter but more regularly defined ignition delays.

1.2 Ignition delay VS pressure

The effect of pressure on the Z-curve is investigated by repeating the ignition-delay calculation for several pressure levels. For a fixed equivalence ratio ϕ, a list of pressures between 1 and 10 atm is defined, and for each pressure P, the zero-dimensional adiabatic constant-volume reactor is initialized successively at all temperatures T_0 in the chosen range.



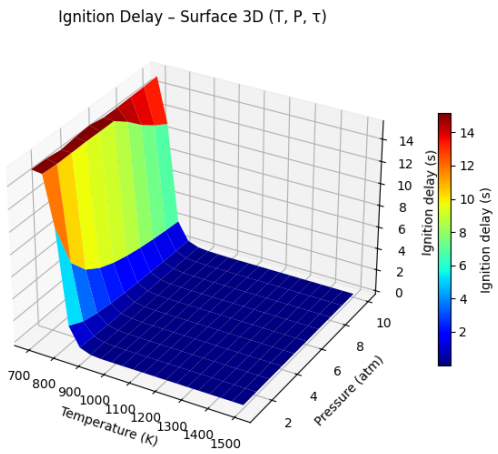
(a) OH based method

(b) Gradient based method

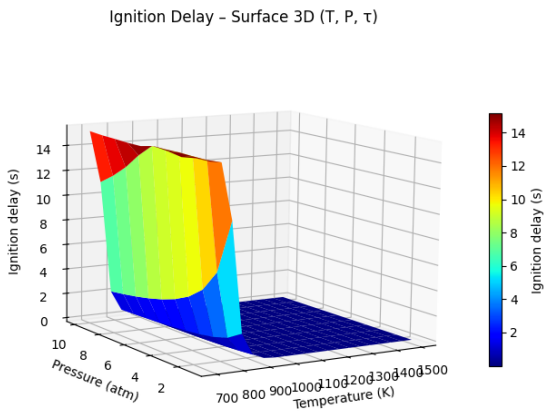
Figure 2: Ignition delay as a function of initial temperature for different initial pressures and fixed equivalence ratio.

The set of curves shows that increasing pressure mainly shifts the Z-curve downwards and slightly towards lower temperatures. At a given T_0 in the low-intermediate range (about 750–1050 K), higher pressures give shorter ignition delays because the larger reactant and third-body concentrations accelerate chain-branching reactions. As a result, the "knee" of the Z-curve occurs at slightly lower temperatures and smaller delay times when the pressure increases, whereas in the high-temperature regime (above ≈ 1100–1200 K) the curves for different pressures become nearly parallel, indicating a weaker pressure sensitivity while the overall Z-shape is preserved.

There is also a discrepancy between the two methods here: the gradient approach provides results consistent with the literature and with expected behavior, since it follows the overall exothermic response, whereas the OH-based criterion shows peaks at certain pressures due to its high sensitivity linked to the tracking of an isolated radical, whose concentration is governed by complex radical dynamics.



(a)



(b)

Figure 3: Ignition delay as a function of initial temperature for different initial pressures and fixed equivalence ratio (two viewing angles).

Figure 3 shows that temperature is the main factor driving the response, while pressure plays a more secondary role. At low temperatures, we see that an increase in P_0 slightly reduces the ignition delay, indicating slightly accelerated kinetics, whereas at higher temperatures the influence of pressure becomes very limited and the curves remain generally smooth, with no marked non-monotonic variations in the range studied.

2 Laminar Flame Speed

2.1 Calculation of the laminar flame speed of a stoechiometric $H_2 - O_2$ flame.

2.1.1 Stoichiometric H_2–O_2 laminar flame speed

For the stoichiometric premixed H_2–O_2 flame considered here, the one-dimensional Cantera calculation gives a laminar flame speed of approximately $S_L \approx 9.8$ m/s using the Multicomponent model and $S_L \approx 10.218$ m/s using the mixture-averaged model. The difference is that the mixture-averaged model considers that each species diffuses in an average mixture, without taking into account the detailed cross-flows between species. This contrasts with the multicomponent model, which considers that the diffusive flows of each species depend on the gradients of the others, which modifies the structure of the flame and makes it more reliable.

If nitrogen is added together with oxygen (i.e. replacing pure H_2–O_2 by an H_2–air mixture), the laminar flame speed is expected to decrease markedly. The presence of N_2 mainly acts as an inert diluent: it increases the overall heat capacity of the mixture, lowers the adiabatic flame temperature, and reduces thermal diffusivity, all of which slow down the flame and lead to laminar burning velocities that are typically a factor of 2–3 lower than in the corresponding undiluted H_2–O_2 case.

2.2 Calculation of the laminar flame speed of an H_2/CH_4 blend as a function of the H_2 percentage.

2.2.1 Equations

For a stoichiometric reaction between H_2 and CH_4 with oxygen, the chemical combustion equations are written as follows:
Hydrogen combustion :

$$\mathrm{H_2} + \frac{1}{2}\mathrm{O_2} \longrightarrow \mathrm{H_2O} \tag{3}$$

Methane combustion :

$$\mathrm{CH_4} + 2\,\mathrm{O_2} \longrightarrow 2\,\mathrm{H_2O} + \mathrm{CO_2} \tag{4}$$

H_2/CH_4 mixture :

$$X_{CH_4}CH4 + X_{H_2}H2$$

$$X_{CH_4} + X_{H_2} = 1 \tag{5}$$

This allows us to calculate the mole number of oxygen we need, and from there, it is easy to find the mole number of nitrogen, so we have:

$$n_{O_2} = \frac{1}{2}X_{H_2} + 2X_{CH_4} \tag{6}$$

Air composition :

$$Air = O_2 + 3.76N_2$$

$$n_{N_2} = 3.76 n_{O_2} \tag{7}$$

The molar fraction is used because, in the gas mixture, it is directly equivalent to the hydrogen volume fraction, allowing the H_2 content to be defined without accounting for the large difference in molar masses between H_2 and CH_4

2.2.2 Parameters influencing the laminar flame speed

The code uses the following process: The laminar flame speed of stoichiometric H_2/CH_4–air mixtures is evaluated using a one-dimensional freely propagating premixed flame for three unburnt temperatures (T_0 = 300, 600 and 900 K). For each temperature, the fuel is defined as a binary mixture of H_2 and CH_4 with the H_2 molar fraction varying from 0 to 1, while the amount of O_2 and N_2 is adjusted to keep the global equivalence ratio at $\phi = 1$.For every composition, a `FreeFlame` calculation is performed and the laminar flame speed S_L is obtained from the inlet gas velocity.

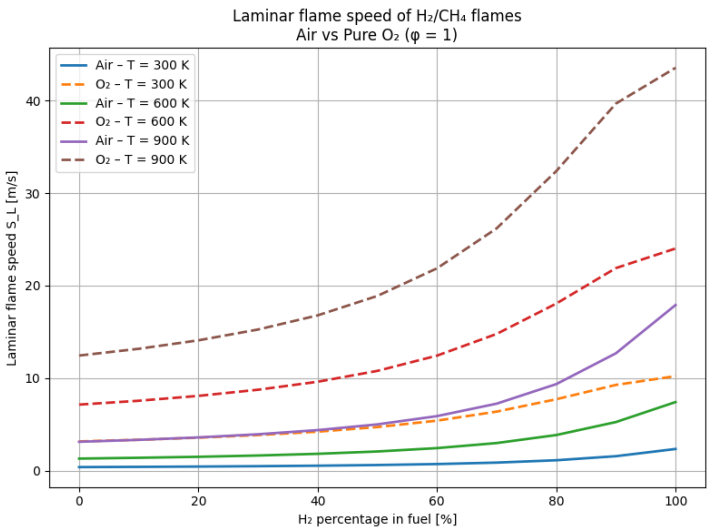


Figure 4: Laminar flame speed of a stoichiometric H_2/CH_4–air mixture as a function of hydrogen percentage in the fuel for different unburnt temperatures.

Figure (4) shows that, for a given initial temperature T_0, the laminar flame speed increases monotonically with hydrogen fraction in the fuel. Even small additions of H_2 to CH_4 significantly enhance S_L, with a sharp increase as the mixture approaches pure hydrogen.

This trend is consistent with experimental and numerical studies of H_2/CH_4–air flames and results from both thermal and chemical effects. Hydrogen enrichment increases the adiabatic flame temperature and promotes chain-branching reactions producing highly reactive H, O, and OH radicals, thereby accelerating the overall reaction rate. This effect is amplified at higher unburnt temperatures, explaining the larger spread in flame speeds observed at T_0 = 900 K compared to 300 K.

For reactions with pure oxygen, the influence of nitrogen dilution is clearly observed: the presence of N_2 significantly reduces the laminar flame speed by lowering the flame temperature, highlighting its strong diluting effect.

3 Prediction of NOx emission for a Perfectly Stirred Reactor (PSR)

3.1 Hypothesis

The combustion system is modeled as a *Perfectly Stirred Reactor* (PSR), also referred to as a *Well-Stirred Reactor*. This modeling approach relies on the following fundamental assumptions:

- **Perfect mixing:** The reactor is assumed to be perfectly mixed, meaning that temperature, pressure, and species mass fractions are spatially uniform throughout the reactor volume. Consequently, the outlet composition is identical to the composition inside the reactor.
- **Steady-state operation:** The reactor operates under steady-state conditions. All temporal derivatives of mass, species, and energy balances are neglected.
- **Ideal gas behavior:** The gas mixture is assumed to obey the ideal gas law. This assumption is justified for H_2–O_2 combustion systems operating at high temperatures and moderate pressures, where intermolecular forces are negligible and the mean free path is large compared to molecular dimensions.
- **Constant pressure:** The pressure inside the reactor is considered constant, which is consistent with typical PSR modeling of combustion systems.
- **Finite-rate chemical kinetics:** Chemical reactions are governed by detailed chemical kinetics. NO_x formation is therefore controlled by temperature, residence time, and equivalence ratio through thermal and kinetic pathways.

The PSR model is particularly suitable for studying NO_x formation because NO_x chemistry is highly sensitive to temperature (Zeldovich mechanism) and chemical residence time, both of which are directly controlled parameters in a PSR.

3.2 Prediction of NO_x emission

Figure 5 presents the total NO_x concentration ($NO + NO_2 + N_2O$) as a function of residence time for different equivalence ratios ϕ. The residence time τ represents the characteristic chemical time available for NO_x formation.

For all equivalence ratios, NO_x emissions increase with residence time at short times, as longer residence times enhance high-temperature reactions dominated by the thermal (Zeldovich) mechanism. At very short residence times ($\tau < 10^{-5}$ s), incomplete chemistry limits NO_x formation.

NO_x reaches a maximum near stoichiometric conditions ($\phi \approx 1$) due to the highest flame temperatures. Lean mixtures ($\phi < 1$) produce less NO_x because of lower temperatures, while rich mixtures ($\phi > 1$) exhibit reduced NO_x formation due to limited oxygen availability.

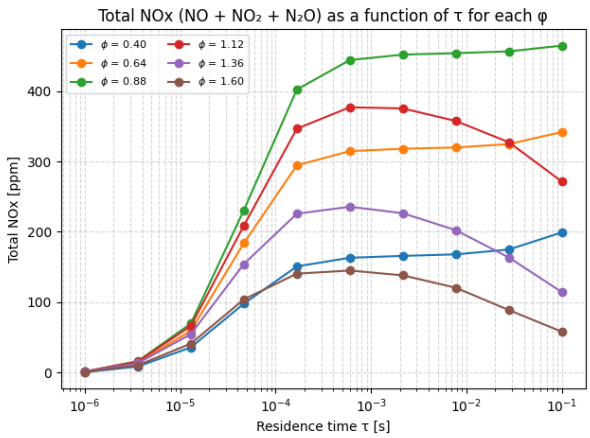
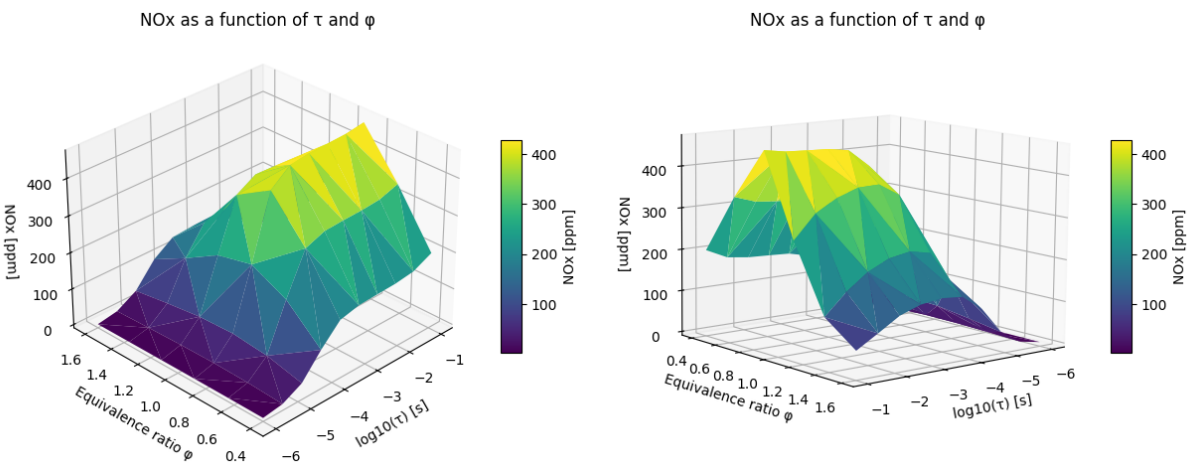


Figure 5: Total NO_x concentration as a function of residence time for different equivalence ratios ϕ in a PSR.

Figures 6a and 6b show the same NO_x data represented as three-dimensional surfaces, highlighting the combined effects of residence time and equivalence ratio.



(a) 3D view of NO_x concentration as a function of residence time and equivalence ratio.

(b) Alternative 3D perspective of NO_x surface emphasizing gradients.

Figure 6: Three-dimensional representation of NO_x emissions in a PSR as a function of residence time and equivalence ratio.

The three-dimensional surfaces clearly illustrate the strong coupling between residence time and equivalence ratio. NO_x formation exhibits a pronounced peak at intermediate residence times and near-stoichiometric conditions. At long residence times, NO_x levels tend to stabilize or slightly decrease due to the approach toward chemical equilibrium and possible NO reburning mechanisms.

Overall, the PSR model successfully captures the fundamental trends of NO_x formation in H_2–O_2 combustion, demonstrating that NO_x emissions are maximized at high temperatures, sufficient residence times, and equivalence ratios close to unity.

Conclusion

The present study has provided a comprehensive analysis of the main physicochemical aspects governing hydrogen and hydrogen–methane combustion, combining detailed kinetic modeling with thermodynamic and transport considerations. Through the systematic use of Cantera simulations, ignition delay, laminar flame speed, and NOx formation were all investigated under controlled conditions representative of practical combustion environments.

The ignition-delay analysis revealed the characteristic Z-shaped behavior of hydrogen–oxygen mixtures, which stems from the competition between chain-branching and chain-termination reactions. The distinct temperature regimes identified demonstrated the crucial role of pressure and temperature in controlling radical kinetics and auto-ignition phenomena, in line with trends reported in the literature for hydrogen-based mixtures.

The laminar flame speed computations further highlighted the high reactivity and diffusivity of hydrogen compared to methane. The monotonic increase of flame speed with hydrogen fraction emphasizes the significant potential of hydrogen enrichment for enhancing flame stability and combustion efficiency, consistent with previous experimental and numerical studies of H_2/CH_4–air flames. Additionally, the observed sensitivity to the unburnt gas temperature and the diluting effect of nitrogen underline the importance of both thermal and transport properties in laminar flame propagation.

The PSR modeling of NOx emissions allowed a global understanding of pollutant formation mechanisms in hydrogen combustion. Results showed that NOx emissions reach a maximum under near-stoichiometric conditions and moderate residence times, where high temperatures and oxygen availability promote thermal NO formation via the Zeldovich mechanism, in agreement with recent NOx studies in hydrogen flames. At rich or lean conditions, NOx production decreases due to either oxygen limitation or reduced flame temperature.

Altogether, this work demonstrates the capacity of detailed chemical-kinetic modeling to predict key combustion parameters and pollutant trends for hydrogen-based fuels. The findings confirm hydrogen's strong potential as a low-carbon energy vector, while highlighting that careful control of equivalence ratio, mixing, and residence time remains essential for limiting NOx emissions in future clean and efficient hydrogen combustion systems.